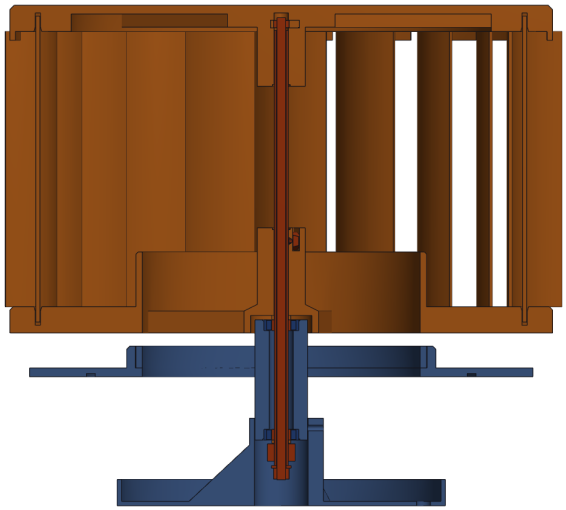
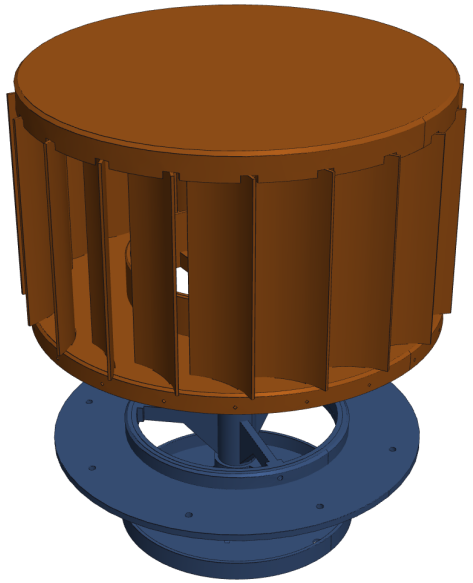
SHEET 3 - ROTOR vs STATIONARY: what turns, what is fixed to the roof, and how each attaches

SECTION - ORANGE rotating / BLUE stationary



Rotor (orange) lifted 70 mm off the stationary support (blue)



ROTATING part (orange)	Attaches to	By
Upper hub (closed top)	shaft + blades	M4 tie rod in tube -> captive nut; pulled onto shoulders
Lower hub (closed bottom)	shaft	1x M4 set screw in nut slot -> shaft dimple
20 blades	both hubs	tabs trapped in blind pockets (no fasteners)
Shaft 1/4 in TUBE + tie rod	bearing INNER races	nyloc nut + washer under the tube end
Top thrust washer	upper bearing inner race	loose on shaft, under lower-hub cup
Shaft collar + bottom washer	lower bearing inner race	collar set screw (uplift stop, 0.2 endplay)

STATIONARY part (blue)	Attaches to	By
Roof flash adapter	roof deck	8x #8 SS screws + sealant bead in groove
Bearing support (ring/spokes/tower)	adapter collar counterbore	rests on step; 3x M3 radial screws
2x R4-2RS bearing OUTER races	bearing support tower	pressed into seats (fit per coupon)
Rain tray (optional)	tower bottom (below roof)	slip-on standpipe + 1x M3 clamp screw

NO stationary part is fastened to either rotating hub. The only paths between rotor and stator are the two ball bearings. Minimum running clearances (from CAD): 6.0 mm lower hub to collar/support ring, 2.0 mm radial labyrinth (cup vs tower), 11.5 mm air channel blade to blade, >13 mm blade to any stationary part.
Service: remove 3 M3 screws -> the whole rotor + bearing cartridge lifts out of the roof adapter.

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Rotor vs stationary

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

SHEET 1/1 Source: build123d parametric model + MASTER_PARAMETERS.json